GREEEN tallgrass prairie restoration: Seeded richness ~ Seeding time * Herbicide pre-treatment (Lux Arbor)

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> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(lme4)
library(glmmTMB)
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    herbicide = col_factor(levels = c("0", "1"), ordered = FALSE),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "Lux Arbor",
    richness_type == "seeded_richness",
    treatment_id %in% c("1", "2", "3", "4")
  ) %>%
  mutate(
    treatment = str_c(seeding_time, herbicide, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  mutate(y = richness_1qm + richness_25qm)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##    upper     mean    lower
## 6.972522 6.258741 5.544960
```

```r
median(sites$y)
```

```
## [1] 7
```

```r
sd(sites$y)
```

```
## [1] 4.317852
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##  5% 95%
##   0  12
```

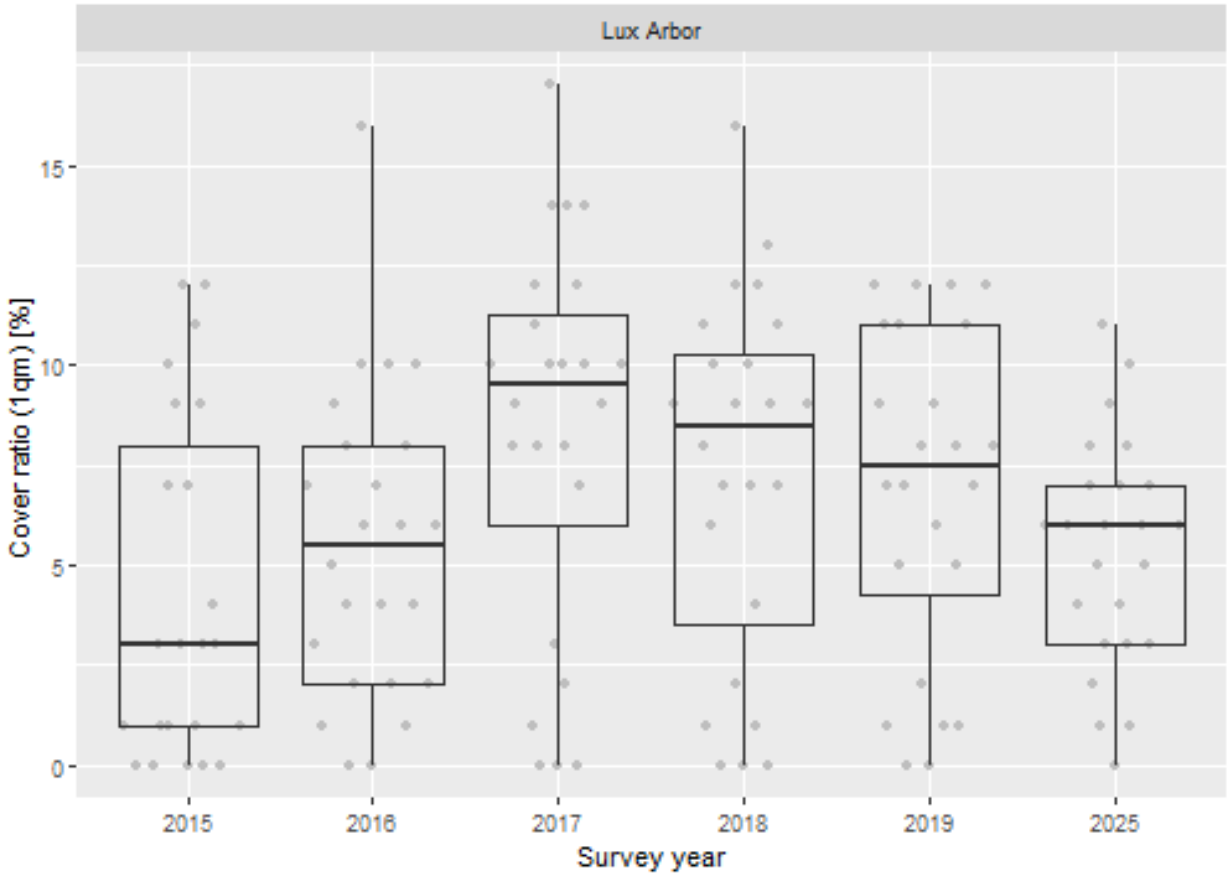
```r
sites %>% count(site, year)
```

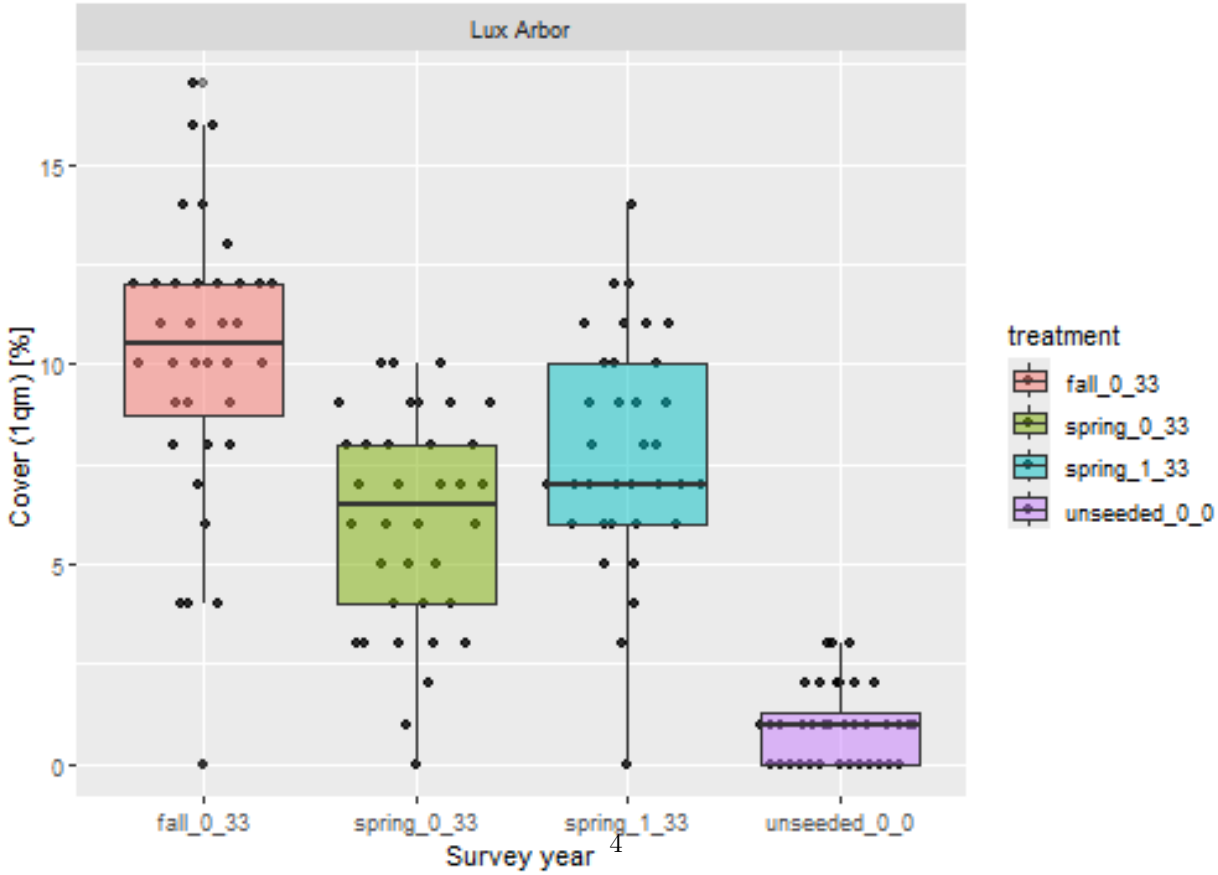
```
## # A tibble: 6 x 3
##   site      year      n
##   <chr>     <fct> <int>
## 1 Lux Arbor 2015     23
## 2 Lux Arbor 2016     24
## 3 Lux Arbor 2017     24
## 4 Lux Arbor 2018     24
## 5 Lux Arbor 2019     24
## 6 Lux Arbor 2025     24
```

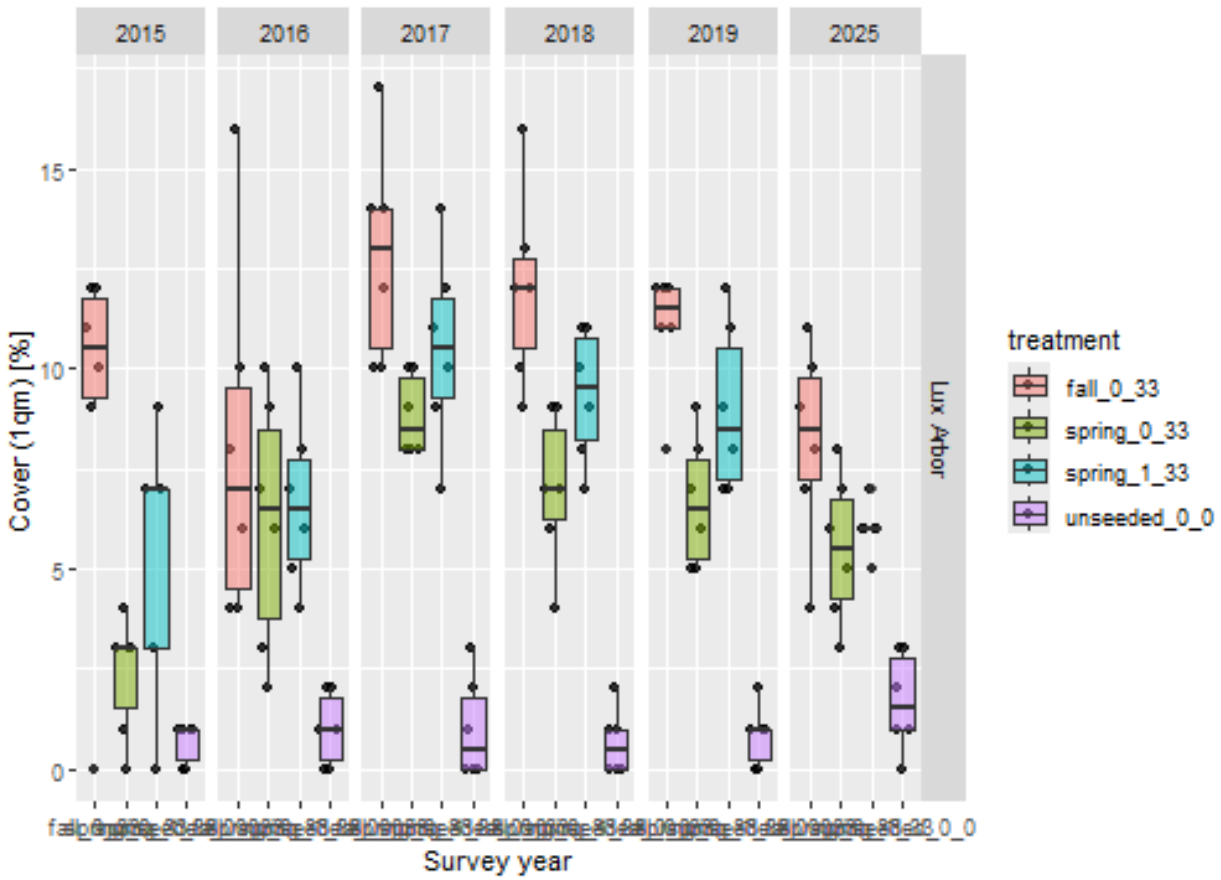
```r
sites %>% count(treatment)
```

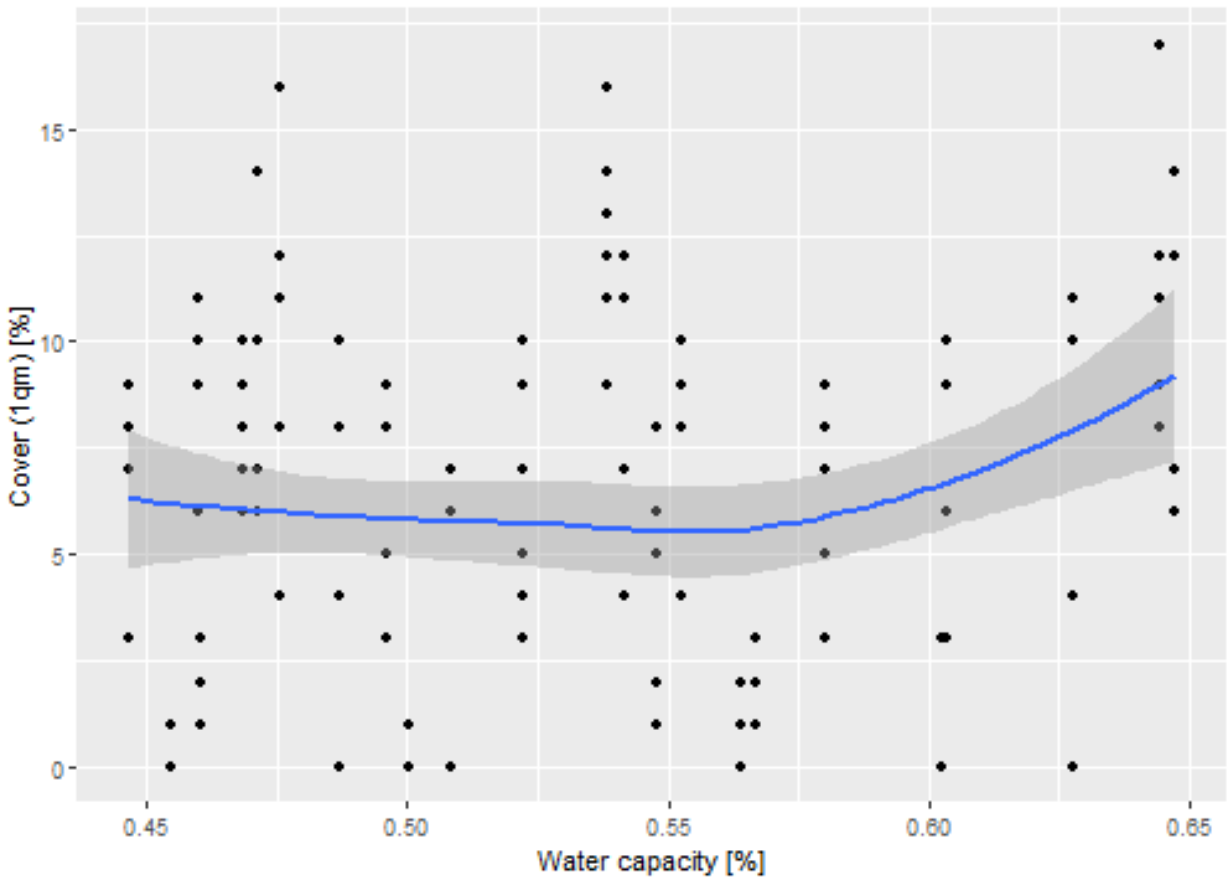
```
## # A tibble: 4 x 2
##   treatment        n
##   <fct>        <int>
## 1 fall_0_33       36
## 2 spring_0_33     36
## 3 spring_1_33     35
## 4 unseeded_0_0    36
```

Graphs of raw data (Step 2, 6, 7)

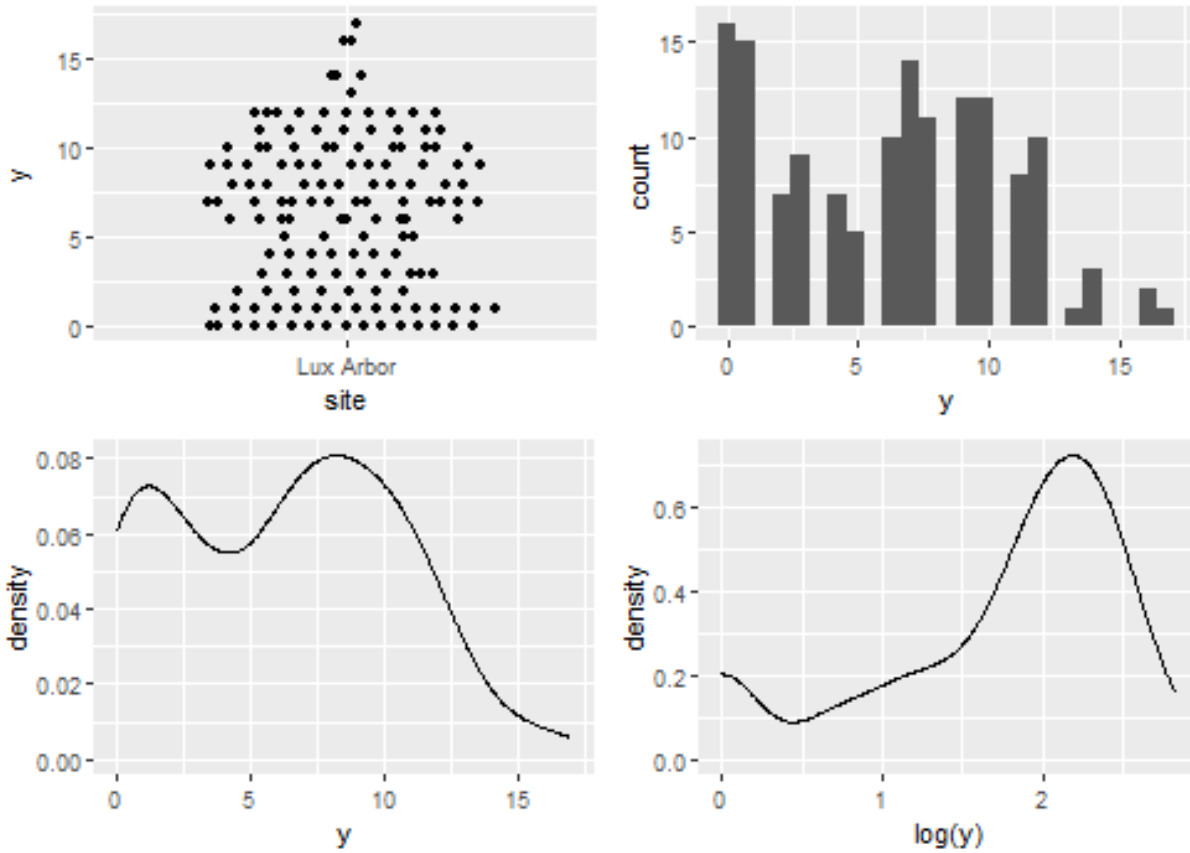





2015
2016
2017
2018
2019
2025
Lux Arbor
Cover (1qm) [%]
15
10
5
0
Survey year
treatment
fall_0_33
spring_0_33
spring_1_33
unseeded_0_0


Cover (1qm) [%]
15
10
5
0
0.45
0.50
0.55
0.60
0.65
Water capacity [%]

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_richness_timing_herbicide_lux_1.Rdata"))
load(file = here("outputs", "models", "model_richness_timing_herbicide_lux_2.Rdata"))
m_1 <- m1
m_2 <- m2
```

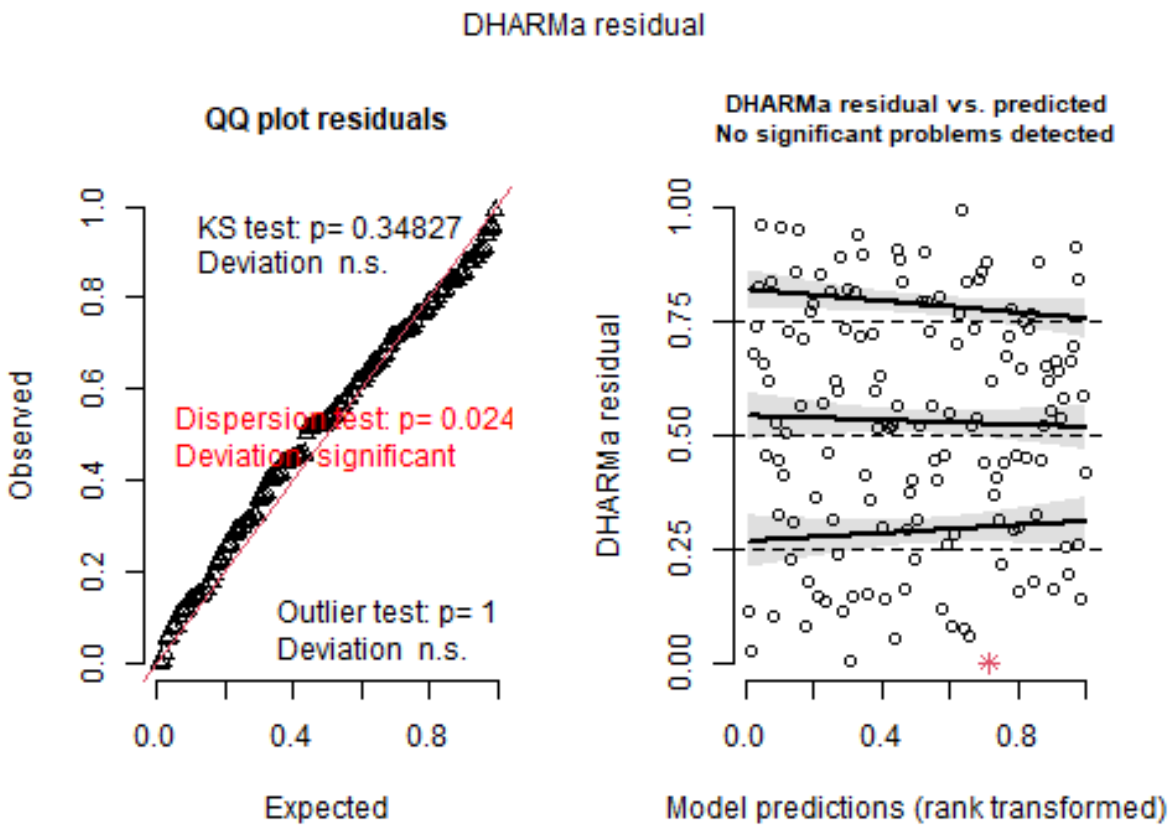
```
formula(m_1)
## y ~ treatment * year + water_cap
formula(m_2)
```

```
## y ~ treatment * year + water_cap
# m_1@call
# m_2@call
```

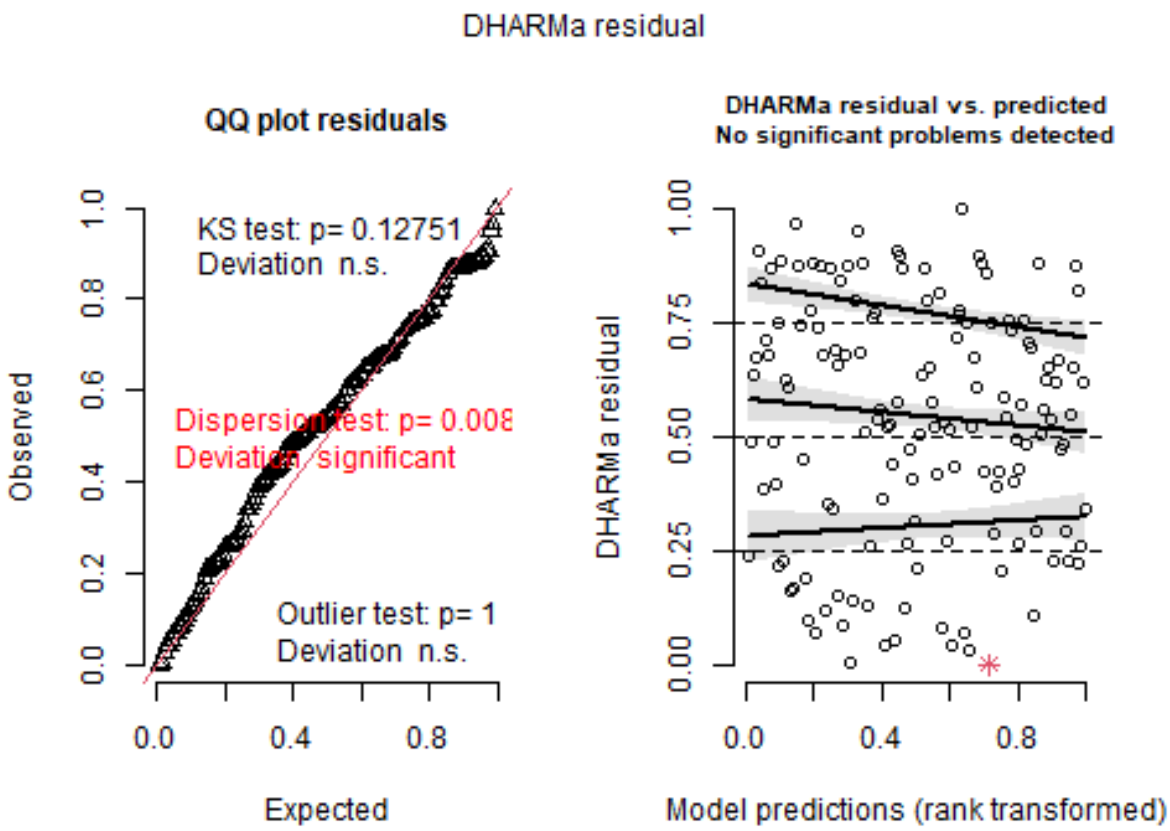
Model check

DHARMa

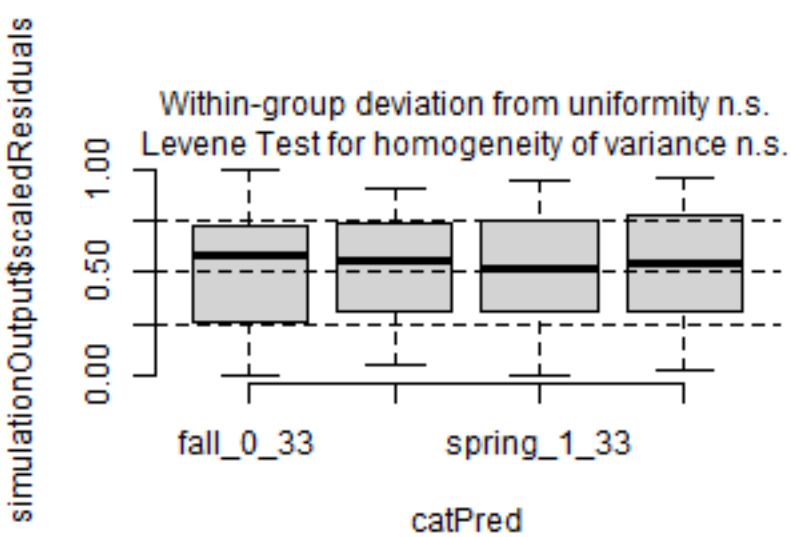
```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



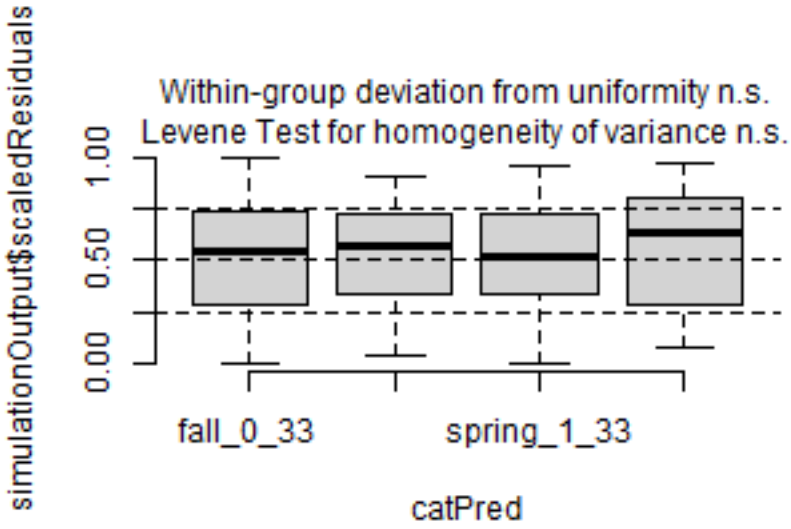

```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



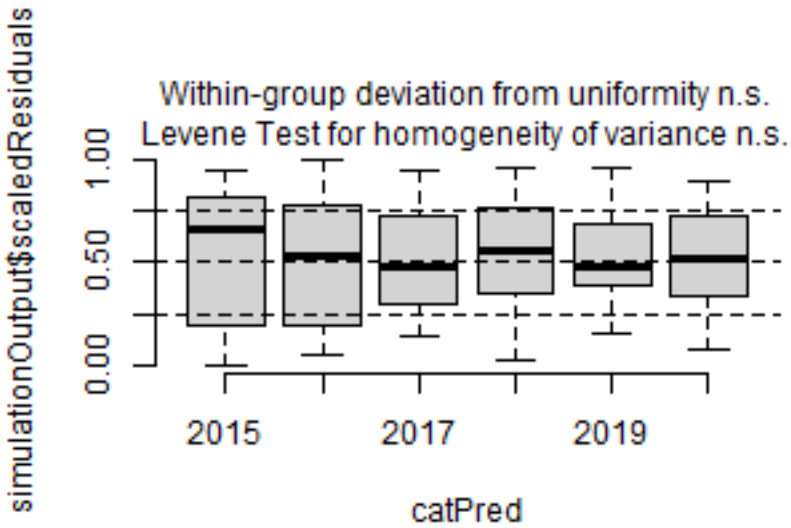

```
plotResiduals(simulation_output_1$scaledResiduals, sites$treatment)
```



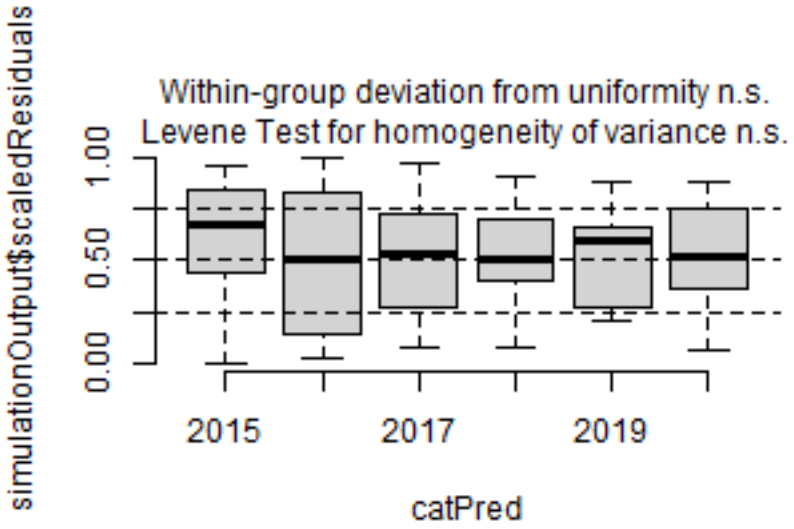

```
plotResiduals(simulation_output_2$scaledResiduals, sites$treatment)
```



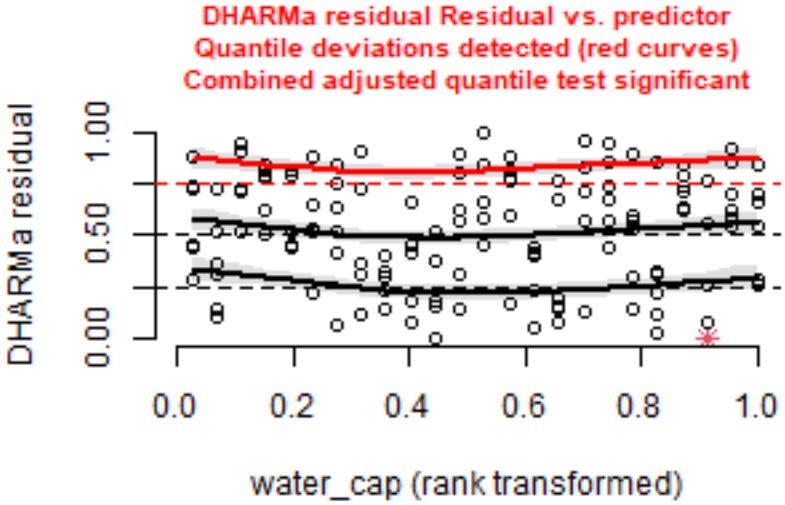

```
plotResiduals(simulation_output_1$scaledResiduals, sites$year)
```



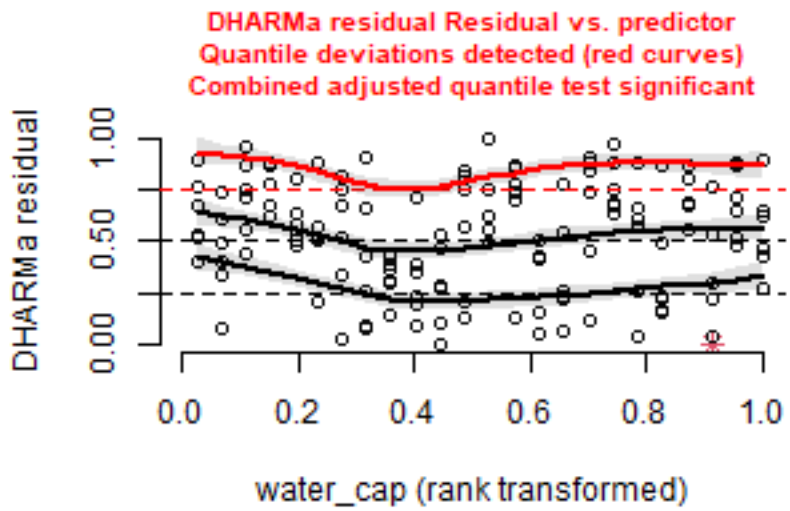

```
plotResiduals(simulation_output_2$scaledResiduals, sites$year)
```




```
plotResiduals(simulation_output_1$scaledResiduals, sites$water_cap)
```

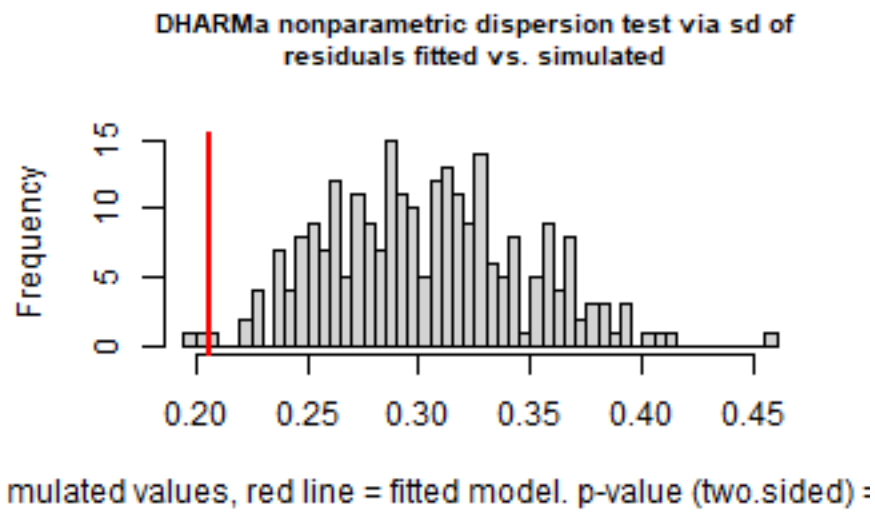



```
plotResiduals(simulation_output_2$scaledResiduals, sites$water_cap)
```

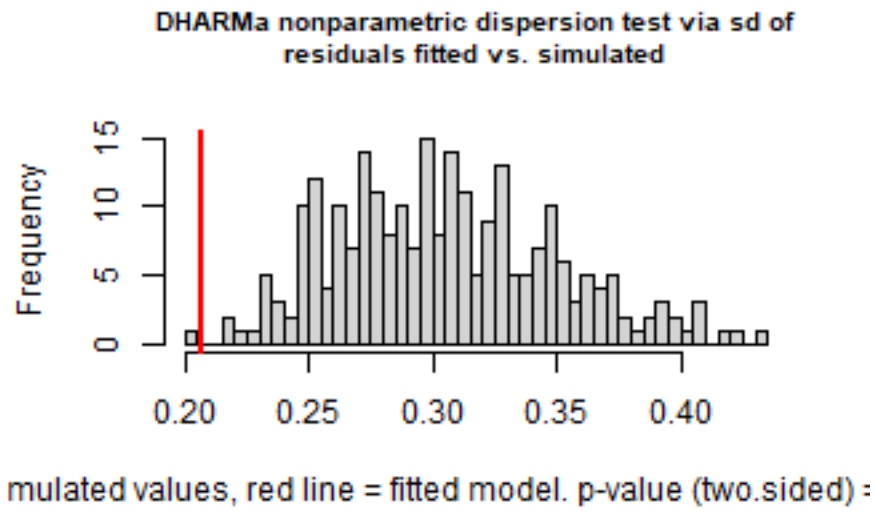


Test for overdispersion

```
testDispersion(simulation_output_1)
```




```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.67945, p-value = 0.024
## alternative hypothesis: two.sided
testDispersion(simulation_output_2)
```




```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.6765, p-value = 0.008
## alternative hypothesis: two.sided
```

Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##                        GVIF Df GVIF^(1/(2*Df))
## treatment       1132.895733  3        3.228729
## year              82.614086  5        1.554911
## water_cap          1.508179  1        1.228080
## treatment:year 53603.785919 15        1.437606
```

```
# car::vif(m_2)
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
## Warning: the null model is only correct if all the variables it uses are identical
## to those used in fitting the original model.
##                 R2m       R2c
## delta     0.8590361 0.8590361
## lognormal 0.8678792 0.8678792
## trigamma  0.8489689 0.8489689
MuMIn::r.squaredGLMM(m_2)
```

```
## Warning: the null model is only correct if all the variables it uses are identical
## to those used in fitting the original model.
##                 R2m       R2c
## delta     0.8590355 0.8590355
## lognormal 0.8678787 0.8678787
## trigamma  0.8489683 0.8489683
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df     AICc
## m_1 25 644.7985
## m_2 26       NA
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)
```

```
## Analysis of Deviance Table (Type III tests)
##
## Response: y
##                LR Chisq Df Pr(>Chisq)
## treatment        59.829  3  6.394e-13 ***
## year             12.505  5    0.02849 *
## water_cap         0.456  1    0.49929
## treatment:year   18.074 15    0.25877
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

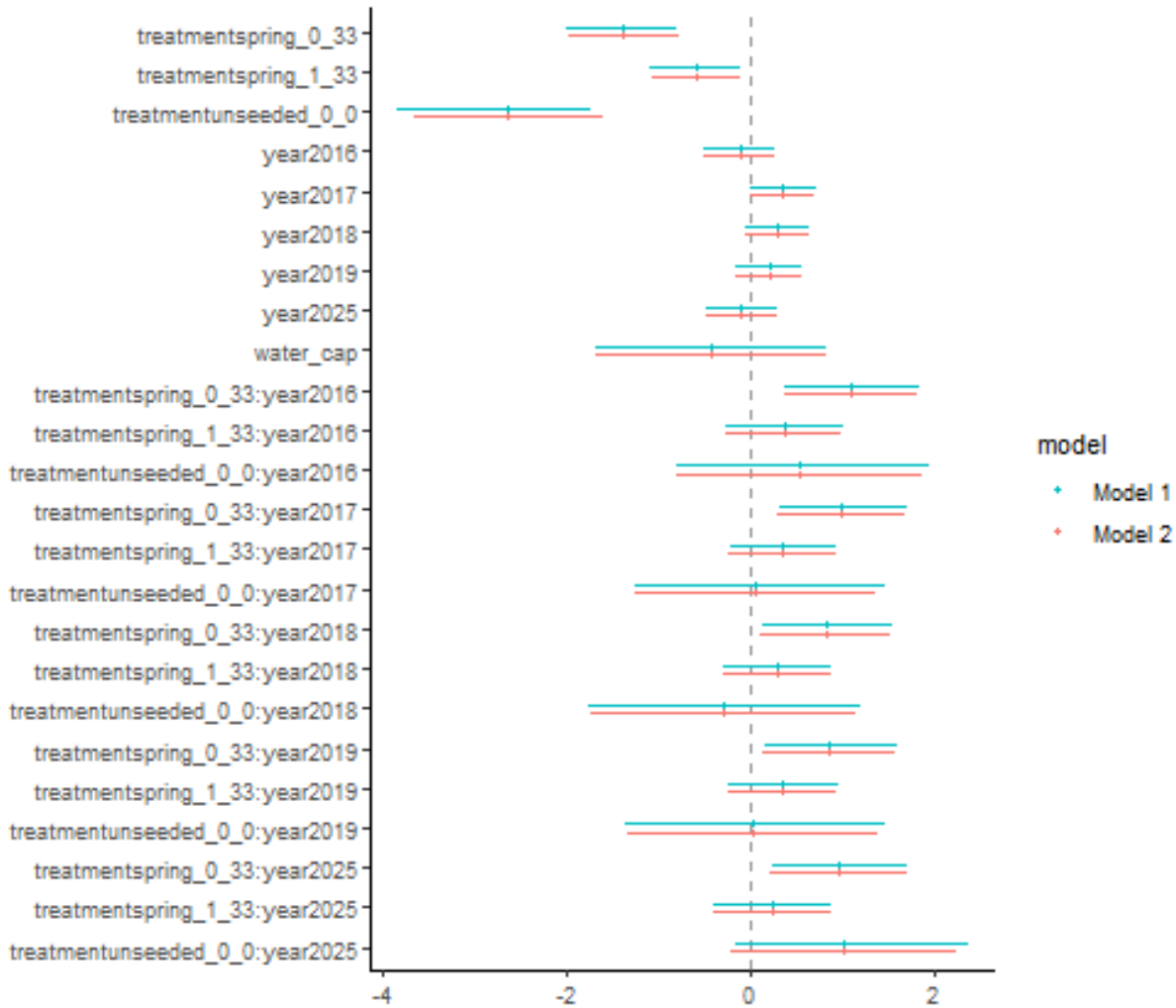
```
summary(m_1)
```

```
##
## Call:
## glm(formula = y ~ treatment * year + water_cap, family = poisson(link = "log"),
##     data = sites)
##
## Coefficients:
##                                Estimate Std. Error z value Pr(>|z|)
## (Intercept)                     2.44755    0.39380   6.215 5.13e-10 ***
## treatmentspring_0_33           -1.37129    0.30155  -4.547 5.43e-06 ***
## treatmentspring_1_33           -0.58538    0.24473  -2.392  0.01676 *
## treatmentunseeded_0_0          -2.62679    0.51940  -5.057 4.25e-07 ***
## year2016                       -0.11778    0.19837  -0.594  0.55268
## year2017                        0.35482    0.17750   1.999  0.04561 *
## year2018                        0.28768    0.18002   1.598  0.11003
## year2019                        0.20067    0.18349   1.094  0.27413
## year2025                       -0.09716    0.19730  -0.492  0.62239
## water_cap                      -0.43160    0.63814  -0.676  0.49882
## treatmentspring_0_33:year2016   1.08964    0.37122   2.935  0.00333 **
## treatmentspring_1_33:year2016   0.36457    0.32065   1.137  0.25556
```

```
## treatmentunseeded_0_0:year2016  0.52325    0.67529   0.775  0.43843
## treatmentspring_0_33:year2017   0.97641    0.34900   2.798  0.00515 **
## treatmentspring_1_33:year2017   0.34622    0.29300   1.182  0.23734
## treatmentunseeded_0_0:year2017  0.05064    0.66945   0.076  0.93970
## treatmentspring_0_33:year2018   0.81093    0.35728   2.270  0.02322 *
## treatmentspring_1_33:year2018   0.29558    0.29788   0.992  0.32106
## treatmentunseeded_0_0:year2018 -0.28768    0.72966  -0.394  0.69338
## treatmentspring_0_33:year2019   0.84915    0.36069   2.354  0.01856 *
## treatmentspring_1_33:year2019   0.34622    0.30109   1.150  0.25019
## treatmentunseeded_0_0:year2019  0.02247    0.69546   0.032  0.97422
## treatmentspring_0_33:year2025   0.95461    0.37504   2.545  0.01092 *
## treatmentspring_1_33:year2025   0.23859    0.32430   0.736  0.46191
## treatmentunseeded_0_0:year2025  1.01345    0.62364   1.625  0.10415
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 524.18  on 142  degrees of freedom
## Residual deviance: 127.29  on 118  degrees of freedom
## AIC: 633.69
##
## Number of Fisher Scoring iterations: 5
```

Forest plot

```r
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

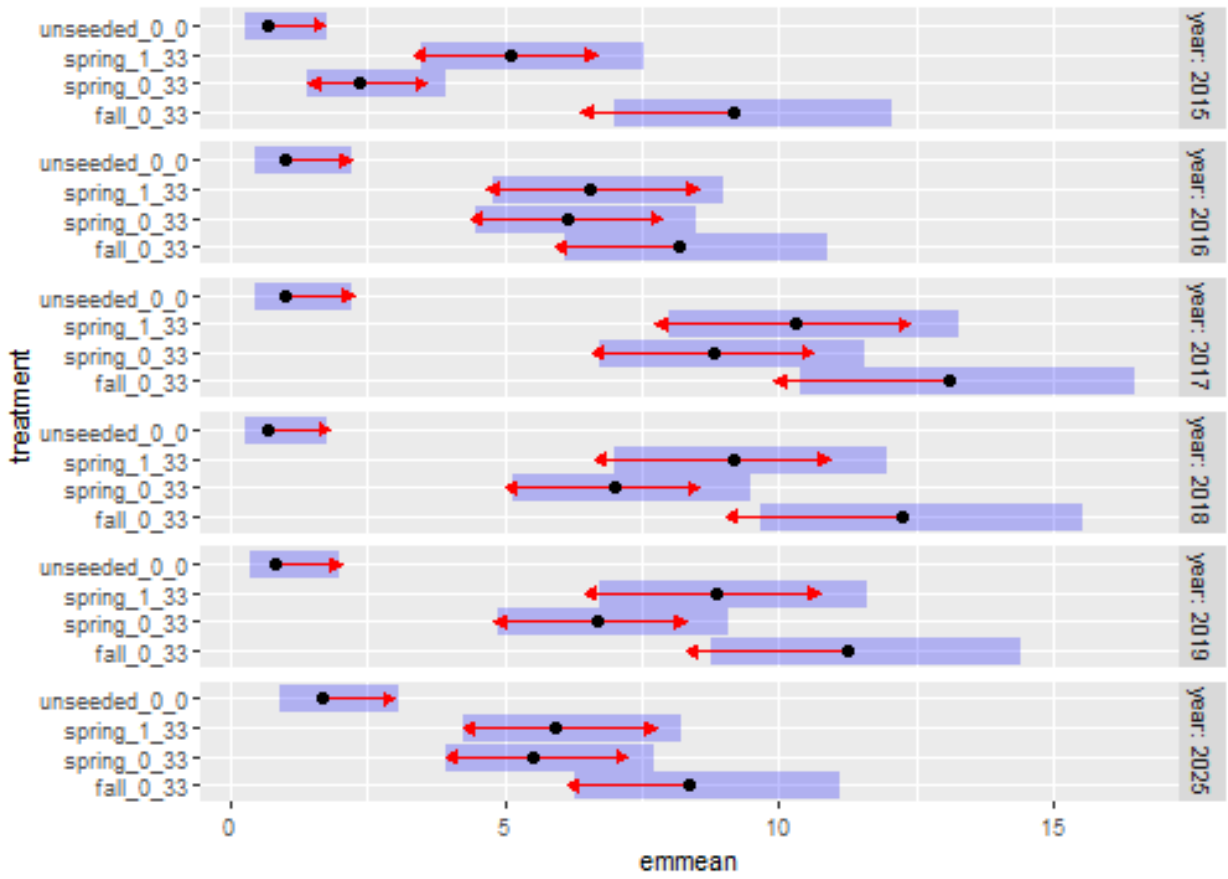
```
(emm <- emmeans(
  m_1,
  revpairwise ~ treatment | year,
  type = "response"
  ))
```

```
## $emmeans
## year = 2015:
##  treatment      rate    SE  df asymp.LCL asymp.UCL
##  fall_0_33     9.187 1.280 Inf     6.992     12.07
##  spring_0_33   2.331 0.623 Inf     1.381      3.94
##  spring_1_33   5.116 1.010 Inf     3.473      7.54
##  unseeded_0_0  0.664 0.332 Inf     0.249      1.77
##
## year = 2016:
##  treatment      rate    SE  df asymp.LCL asymp.UCL
```

```
##  fall_0_33     8.166 1.200 Inf     6.117     10.90
##  spring_0_33   6.162 1.010 Inf     4.464      8.50
##  spring_1_33   6.548 1.050 Inf     4.782      8.97
##  unseeded_0_0  0.996 0.407 Inf     0.448      2.22
##
## year = 2017:
##  treatment      rate    SE  df asymp.LCL asymp.UCL
##  fall_0_33    13.100 1.540 Inf    10.399     16.50
##  spring_0_33   8.826 1.210 Inf     6.743     11.55
##  spring_1_33  10.313 1.330 Inf     8.013     13.27
##  unseeded_0_0  0.996 0.407 Inf     0.448      2.22
##
## year = 2018:
##  treatment      rate    SE  df asymp.LCL asymp.UCL
##  fall_0_33    12.249 1.490 Inf     9.652     15.54
##  spring_0_33   6.994 1.080 Inf     5.169      9.46
##  spring_1_33   9.167 1.250 Inf     7.019     11.97
##  unseeded_0_0  0.664 0.332 Inf     0.249      1.77
##
## year = 2019:
##  treatment      rate    SE  df asymp.LCL asymp.UCL
##  fall_0_33    11.228 1.420 Inf     8.760     14.39
##  spring_0_33   6.661 1.050 Inf     4.886      9.08
##  spring_1_33   8.840 1.230 Inf     6.736     11.60
##  unseeded_0_0  0.830 0.371 Inf     0.346      2.00
##
## year = 2025:
##  treatment      rate    SE  df asymp.LCL asymp.UCL
##  fall_0_33     8.336 1.220 Inf     6.263     11.10
##  spring_0_33   5.496 0.957 Inf     3.907      7.73
##  spring_1_33   5.893 0.995 Inf     4.233      8.20
##  unseeded_0_0  1.661 0.525 Inf     0.893      3.09
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
## year = 2015:
##  contrast                    ratio     SE  df null z.ratio p.value
##  spring_0_33 / fall_0_33    0.2538 0.0765 Inf    1  -4.547  <.0001
##  spring_1_33 / fall_0_33    0.5569 0.1360 Inf    1  -2.392  0.0786
##  spring_1_33 / spring_0_33  2.1944 0.7290 Inf    1   2.365  0.0839
##  unseeded_0_0 / fall_0_33   0.0723 0.0376 Inf    1  -5.057  <.0001
##  unseeded_0_0 / spring_0_33 0.2849 0.1620 Inf    1  -2.214  0.1193
##  unseeded_0_0 / spring_1_33 0.1298 0.0698 Inf    1  -3.799  0.0008
##
## year = 2016:
##  contrast                    ratio     SE  df null z.ratio p.value
##  spring_0_33 / fall_0_33    0.7545 0.1670 Inf    1  -1.274  0.5793
##  spring_1_33 / fall_0_33    0.8019 0.1780 Inf    1  -0.997  0.7509
##  spring_1_33 / spring_0_33  1.0627 0.2440 Inf    1   0.265  0.9935
##  unseeded_0_0 / fall_0_33   0.1220 0.0530 Inf    1  -4.842  <.0001
##  unseeded_0_0 / spring_0_33 0.1617 0.0712 Inf    1  -4.140  0.0002
##  unseeded_0_0 / spring_1_33 0.1522 0.0667 Inf    1  -4.296  0.0001
```

```
##
## year = 2017:
##  contrast                    ratio     SE  df null z.ratio p.value
##  spring_0_33 / fall_0_33    0.6738 0.1220 Inf    1  -2.179  0.1291
##  spring_1_33 / fall_0_33    0.7873 0.1410 Inf    1  -1.336  0.5399
##  spring_1_33 / spring_0_33  1.1685 0.2200 Inf    1   0.828  0.8412
##  unseeded_0_0 / fall_0_33   0.0761 0.0324 Inf    1  -6.057  <.0001
##  unseeded_0_0 / spring_0_33 0.1129 0.0486 Inf    1  -5.064  <.0001
##  unseeded_0_0 / spring_1_33 0.0966 0.0413 Inf    1  -5.463  <.0001
##
## year = 2018:
##  contrast                    ratio     SE  df null z.ratio p.value
##  spring_0_33 / fall_0_33    0.5710 0.1120 Inf    1  -2.849  0.0228
##  spring_1_33 / fall_0_33    0.7484 0.1400 Inf    1  -1.551  0.4073
##  spring_1_33 / spring_0_33  1.3107 0.2700 Inf    1   1.316  0.5529
##  unseeded_0_0 / fall_0_33   0.0542 0.0279 Inf    1  -5.660  <.0001
##  unseeded_0_0 / spring_0_33 0.0950 0.0497 Inf    1  -4.499  <.0001
##  unseeded_0_0 / spring_1_33 0.0725 0.0375 Inf    1  -5.067  <.0001
##
## year = 2019:
##  contrast                    ratio     SE  df null z.ratio p.value
##  spring_0_33 / fall_0_33    0.5933 0.1200 Inf    1  -2.574  0.0493
##  spring_1_33 / fall_0_33    0.7873 0.1510 Inf    1  -1.246  0.5976
##  spring_1_33 / spring_0_33  1.3271 0.2790 Inf    1   1.347  0.5330
##  unseeded_0_0 / fall_0_33   0.0740 0.0344 Inf    1  -5.598  <.0001
##  unseeded_0_0 / spring_0_33 0.1247 0.0591 Inf    1  -4.389  0.0001
##  unseeded_0_0 / spring_1_33 0.0939 0.0440 Inf    1  -5.055  <.0001
##
## year = 2025:
##  contrast                    ratio     SE  df null z.ratio p.value
##  spring_0_33 / fall_0_33    0.6592 0.1500 Inf    1  -1.833  0.2580
##  spring_1_33 / fall_0_33    0.7070 0.1600 Inf    1  -1.530  0.4194
##  spring_1_33 / spring_0_33  1.0724 0.2600 Inf    1   0.288  0.9916
##  unseeded_0_0 / fall_0_33   0.1992 0.0695 Inf    1  -4.625  <.0001
##  unseeded_0_0 / spring_0_33 0.3022 0.1090 Inf    1  -3.315  0.0051
##  unseeded_0_0 / spring_1_33 0.2818 0.1010 Inf    1  -3.537  0.0023
##
## P value adjustment: tukey method for comparing a family of 4 estimates
## Tests are performed on the log scale
```

```r
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     glmmTMB_1.1.12   lme4_1.1-37
##  [5] Matrix_1.7-3     patchwork_1.3.1  ggbeeswarm_0.7.2 lubridate_1.9.4
##  [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
```

```
## [13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_3.5.2
## [17] tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         compiler_4.5.1
##  [7] mgcv_1.9-3             vctrs_0.6.5            pkgconfig_2.0.3
## [10] crayon_1.5.3           fastmap_1.2.0          backports_1.5.0
## [13] labeling_0.4.3         utf8_1.2.6             ggstance_0.3.7
## [16] promises_1.3.3         rmarkdown_2.29         tzdb_0.5.0
## [19] nloptr_2.2.1           bit_4.6.0              xfun_0.52
## [22] later_1.4.2            parallel_4.5.1         R6_2.6.1
## [25] gap.datasets_0.0.6     stringi_1.8.7          qgam_2.0.0
## [28] RColorBrewer_1.1-3     car_3.1-3              boot_1.3-31
## [31] numDeriv_2016.8-1.1    estimability_1.5.1     Rcpp_1.1.0
## [34] iterators_1.0.14       knitr_1.50             zoo_1.8-14
## [37] parameters_0.27.0      httpuv_1.6.16          splines_4.5.1
## [40] timechange_0.3.0       tidyselect_1.2.1       rstudioapi_0.17.1
## [43] abind_1.4-8            yaml_2.3.10            MuMIn_1.48.11
## [46] doParallel_1.0.17      TMB_1.9.17             codetools_0.2-20
## [49] lattice_0.22-7         plyr_1.8.9             shiny_1.11.1
## [52] withr_3.0.2            bayestestR_0.16.1      coda_0.19-4.1
## [55] evaluate_1.0.4         marginaleffects_0.28.0 pillar_1.11.0
## [58] gap_1.6                carData_3.0-5          foreach_1.5.2
## [61] stats4_4.5.1           reformulas_0.4.1       insight_1.3.1
## [64] generics_0.1.4         vroom_1.6.5            rprojroot_2.0.4
## [67] hms_1.1.3              scales_1.4.0           minqa_1.2.8
## [70] xtable_1.8-4           glue_1.8.0             tools_4.5.1
## [73] data.table_1.17.8      mvtnorm_1.3-3          grid_4.5.1
## [76] rbibutils_2.3          datawizard_1.1.0       nlme_3.1-168
## [79] Rmisc_1.5.1            performance_0.15.0     beeswarm_0.4.0
## [82] vipor_0.4.7            Formula_1.2-5          cli_3.6.5
## [85] gtable_0.3.6           digest_0.6.37          farver_2.1.2
## [88] htmltools_0.5.8.1      lifecycle_1.0.4        mime_0.13
## [91] bit64_4.6.0-1          dotwhisker_0.8.4       MASS_7.3-65
```